

Assembly reference: what is ready, what is open

companion to *cosh_methodology*, *access_tight_rms_bound*, *ge_offset_shift*, *noise_model_notes*

1 Item A — why the moment pair may be averaged

Status: complete. Theory, closed form and measurement all exist; only the manuscript prose is left.

The general statement. Write the unmodelled part of the onset balance as $\Delta_{\pm}$ and split it, $\Delta_a = (\Delta_+ - \Delta_-)/2$, $\Delta_s = (\Delta_+ + \Delta_-)/2$. *Everything that acts as force $\times$ pivot distance changes sign with the tip direction and therefore lands in Δ_a , so*

$$M_{\text{ff}} = \frac{1}{2}(M_+ + M_-) = \frac{1}{2}(W - f)(l_+ - l_-) + S_{\text{off}} + \Delta_s,$$

in which Δ_a has cancelled identically. The ground-effect thrust channel, a common displacement of the contact resultant and rolling resistance are all instances. Measured: $|\Delta_a| = 54$ mNm and $|\Delta_s| = 37$ mNm at the median, the latter 3.7% of the half-width of the interval $[M_-, M_+]$ the excitation measures directly (1000 mNm median, $8.2\times$ the worst $|\Delta_s|$). (*cosh_methodology*, present text)

The closed form for the offset. With the matched-thrust gains (α_f, α_M) ,

$$\begin{aligned}\hat{p}_{\text{off},s} - p_{\text{off}} &= -\frac{\alpha_M(p_{\text{off}} + s l_s)}{1 + \alpha_M} + \frac{s f l_s (\alpha_M - \alpha_f)}{W(1 + \alpha_M)}, \\ \hat{p}_{\text{off,avg}} - p_{\text{off}} &= -\frac{\alpha_M p_{\text{off}}}{1 + \alpha_M} - \left[\alpha_M - \frac{f(\alpha_M - \alpha_f)}{W} \right] \frac{l_+ - l_-}{2(1 + \alpha_M)}.\end{aligned}$$

The slope in p_{off} is $-\alpha_M/(1 + \alpha_M) = -4.13\%$ — the moment channel alone; the thrust channel only scales the arm-asymmetry constant and vanishes at $\alpha_f = \alpha_M$, where the pair average reduces to Eq. (120). Over the ± 20 mm design box a single direction carries ± 4.1 – 5.8 mm while the pair average stays within ± 0.83 mm. (*matlab/offset_ge_theory.m*, *ge_offset_shift*)

The measurement. Per direction 7.5 / 6.4 mm RMS (pooled 7.0, worst 12.2); paired 1.77 mm RMS (worst 2.96), with per-configuration 95% confidence intervals of 0.37–1.73 mm (median 0.66). Ground effect alone predicts 5.4 / 5.1 $\rightarrow$ 0.48 mm, and 12 of the 20 single-direction estimates fall outside that bound — so the pairing removes more than ground effect: it removes every antisymmetric source at once. (*offset_experimental_split.py*, *fig_offset_experimental*)

The truth it is measured against. Load-cell standard error 0.003–0.64 mm over the ten case-axis measurements, pooled 0.30, median 0.03; datasheet propagation $u_p = l u_f / W = 0.15$ mm. Contributes 2.8% of the variance of the paired identification error and is uncorrelated with it ($r = -0.08$). (*loadcell_uncertainty.py*, *DataSet/loadcell*)

2 Item B — the calibration in the introduction

Status: pipeline settled; prose to write.

The order is worth stating exactly, because it is easy to read backwards. *Stage 1:* per-run free nonlinear least squares (PNLS) over (C_1, C_2, C) ; the per-dataset *medians* of (C_2, K) locate a search centre. *Stage 2:* the ramp-invariance score is minimised on a grid around that centre, fixing (C_2, K) ; these are frozen in *pnls_constants.py*. Then COSH runs with the amplitude pinned at $C_1 = KM$, the baseline at the median, and the onset swept exhaustively. PNLS seeds the calibration; it is not part of the estimator.

The robustness sentence to include. The *cosh_wide* variant minimises the same score over the wide box $[3, 8] \times [0.05, 0.70]$ *with no PNLS centre* and is visually indistinguishable from COSH. This is the one line that forecloses “the result depends on the PNLS seed”.

3 Item C — the Gazebo ablation

Status: open. This is the only item with no material yet.

What it should target. The hardware cannot separate the static ground effect from a contact-resultant shift: a 19 mm inboard shift closes the same threshold gap that the interference model closes. Simulation can set the two independently, so the ablation is worth most if aimed there rather than at re-confirming what the hardware already shows.

run	configuration	what it establishes
A	GE off, lever exact	the pipeline is unbiased: identified offset = truth
B	GE on, lever exact	the gap appears, and its size is testable against the interference model's ≈ 64 mN m residual level
C	GE off, lever shifted 19 mm	the same gap reappears — the degeneracy demonstrated directly rather than argued
D	slow ramp, $u > 1$	optional: exercises the admissibility floor of (9a), $\dot{M} > \bar{\rho}C_2$

A, B and C together let the manuscript say: *the two mechanisms are individually sufficient and jointly unidentifiable at this clearance*, which is stronger than the present argument and costs no hardware.

4 Item D — figures and tables already available

figure	script	carries
<code>fig_estimator_err</code>	<code>estimator_error.figure.py</code>	offset error, six estimators, 95% CI
<code>fig_mcrit_static</code>	<code>mcrit_static.figure.py</code>	thresholds vs three physics levels
<code>fig_mcrit_report</code>	<code>mcrit_report.figure.py</code>	parity plot + physics ladder
<code>fig_offset_experimental</code>	<code>offset_experimental.split.py</code>	per-direction vs paired offsets
<code>fig_kernel_free</code>	<code>kernel_free.bound.py</code>	the residual cap on all 140 runs
<code>fig_sgBars</code>	<code>sg_boundBars.py</code>	the cap with the declared noise constant
<code>fig_design_range</code>	<code>design_range.chart.py</code>	the bound over the design box
<code>fig_noise_spectrum</code>	<code>noise_spectrum.figure.py</code>	the spectral construction behind (19)

Tables come from `mcrit_prediction.csv` (per case/axis/direction) and `mcrit_per_run.csv` (per run), both written by `mcrit_prediction.py`.

5 Numbers to quote, in one place

quantity	value	source
static residual, no GE $\rightarrow$ interference	191 $\rightarrow$ 64 mN m (per-run median)	<code>mcrit_report.figure.py</code>
of which the thrust matching	143 $\rightarrow$ 64 mN m	<code>mcrit_matched_thrust.py</code>
one-sided over-prediction	20/20 $\rightarrow$ 12/20	<code>mcrit_prediction.py</code>
share of the gap explained	81%	<code>cosh_methodology</code>
offset error, single $\rightarrow$ paired	7.0 $\rightarrow$ 1.77 mm RMS	<code>offset_experimental.split.py</code>
load-cell truth uncertainty	0.30 mm pooled (0.64 worst)	<code>loadcell_uncertainty.py</code>
residual cap coverage	140/140, used 0.36–0.41	<code>kernel_free.bound.py</code>
noise constant N_n	$2.31^\circ/\text{s} = 3N_{\text{med}}$	<code>sg_boundBars.py</code>
realised δe_ω	$0.16\text{--}0.17^\circ/\text{s}$	<code>kernel_free.bound.py</code>
$\bar{\rho}$ at the 5° box	1.8–3.3 mN m	<code>design_range.chart.py</code>
GE share of $\bar{\rho}$	2–10%; $6\times$ inflation $\rightarrow$ cap +0.1%	<code>ge_channel.share.py</code>

6 Two cautions carried forward

The dynamic ground effect. The attitude dependence is not resolvable here: the inversion's sensitivity floor is ≈ 90 mN m — the static repeatability — against a model variation of -2.5 to -0.1 mN m/deg over the 5° cap.

The bound route is not a fallback but the tighter statement: unmodelled forcing that the family cannot absorb shows in the residual, capped at 140/140; what the family does absorb shifts M_{crit} by at most $\bar{\rho} = 1.8\text{--}3.3$ mN m through (21). Both branches must be stated together — the first alone invites “the fit absorbed it”.

Attribution language. The interference model *accounts for* the gap; it does not *identify* ground effect, because a 19 mm contact-resultant shift closes the same gap. Keep “the size of the near-ground correction, not its attribution” wherever the static result is stated.